

TectoPower

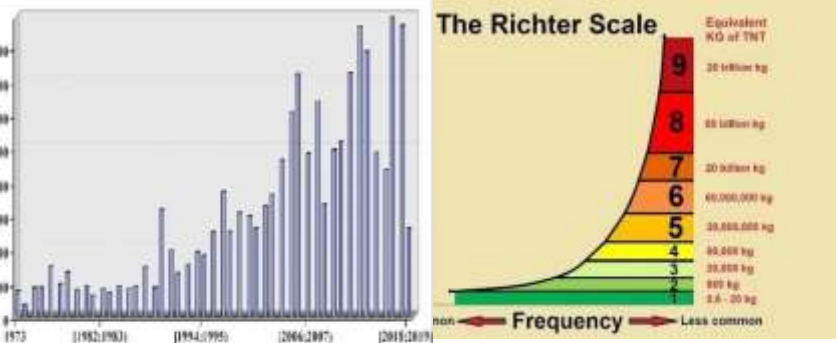
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Abstract

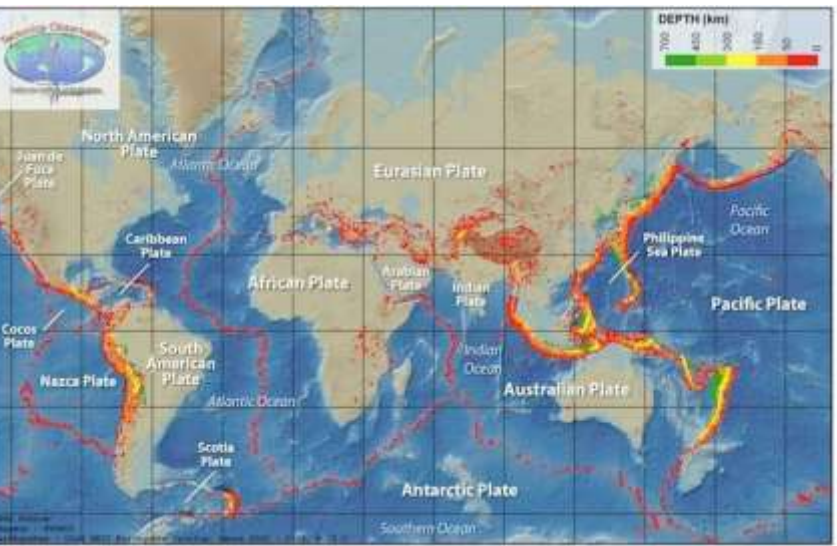
Earthquakes create destructive horizontal forces that threaten structural safety, especially in areas lacking affordable vibration-mitigation technologies. This project presents a low-cost prototype that combines a mechanical damping system with an electromagnetic energy-harvesting unit to demonstrate dual functionality under simulated seismic motion. The simulation platform uses a **DC motor** powered by **two lithium-ion batteries** in a **closed circuit** controlled by a push button. An eccentric metal weight on the motor shaft generates horizontal vibrations that replicate small-scale earthquake activity. Above the shaking base, a spring-supported structure reduces transmitted force, illustrating how simple mechanical elements can limit building displacement. At the same time, vertical motion drives an induction generator composed of a copper coil around an iron core placed between **two U-shaped permanent magnets**. The relative movement produces electrical energy sufficient to power an **LED**, providing clear proof of energy harvesting. This integrated model demonstrates an accessible, scalable approach to sustainable seismic-response technology.

Introduction

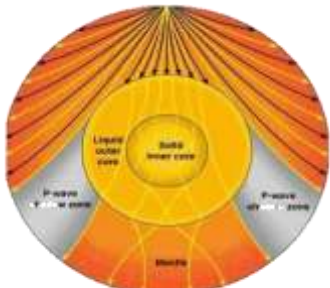
Earthquakes generate destructive horizontal forces that threaten structural safety, particularly in regions lacking affordable vibration-mitigation technologies. They cause severe property damage, injuries, and fatalities, with developing areas being disproportionately affected (Kumar et al., 2020). Many small-scale buildings remain highly vulnerable due to limited access to advanced damping systems. This highlights the urgent need for low-cost, effective solutions to reduce seismic impact while enhancing structural resilience.



Figure(1) shows the high rate of happening the earthquakes over time (Verdon et al., 2024)



Figure(3, 4) illustrate why the earthquakes get happened over time due to **solid inner core** inside the earth and friction of **tectonic plates** to each others (Badawi, 2023).



Background Research

Table (1), represents the prior solutions and in terms of description, advantages, disadvantages.

Prior Solution	Description	Advantages	Disadvantages
Parkfield experiment Roeloffs, E., & Langbein, J. (1994)	Since 1985, Parkfield on the San Andreas Fault has been intensely monitored because six past ~M6 earthquakes occurred at regular intervals. Scientists expected another event before 1993, but it actually happened in 2004.	- Provided one of the best seismic and geodetic datasets ever collected. - Improved understanding of fault behavior and slip processes. - Showed the limits of short-term earthquake prediction.	- The main prediction failed (earthquake came 11 years late). - A 1992 alert was a false alarm. - No reliable short-term precursors were identified.
Seismic Piezoelectric Energy Harvesting Experiment (Xie et al., 2025)	A piezoelectric seismic energy harvester converts ground vibrations from earthquakes into electricity. During experiments using typical seismic waves (e.g., El-Centro wave, PGA = 0.024 g), the device produced 104 V and 11 mW RMS, enough to power small sensors	- Uses natural seismic energy to generate electricity. - Can power small monitoring devices without batteries. - Demonstrates the feasibility of energy harvesting from ground vibrations.	- Very low power output, insufficient for large devices. - Earthquakes are unpredictable, so energy supply is irregular. - Small vibrations produce minimal electricity. - High-quality piezo materials and setup can be costly.

Engineering Goals

How can a dual-function seismic platform be optimized to predict earthquakes using an AI-driven model, harvest electrical energy efficiently via fixed magnets and a vertically oscillating coil, and mitigate destructive forces through spring-based damping, while ensuring affordability and scalability for educational and research.

Efficient Energy Harvesting: Fixed magnets combined with a vertically oscillating coil maximize electromagnetic energy conversion from seismic vibrations, powering LEDs or small devices reliably.

AI-Powered Earthquake Prediction: Integrated machine-learning model analyzes seismic and geophysical data to estimate short-term earthquake probability, providing actionable insights for proactive safety measures and educational demonstrations.

Seismic Force Mitigation: Spring-based damping and stabilized magnet setup absorb and cancel destructive forces, preventing structural damage while maintaining system functionality.

Methodology

Electricity Harvesting Method

Firstly, two **U-shaped magnets** are fixed in parallel, creating a stable magnetic field. A **coiled copper wire** moves **vertically** between them, converting seismic vibrations into electricity via electromagnetic induction. This setup keeps the magnets stationary while maximizing energy harvesting efficiency from the moving coil.

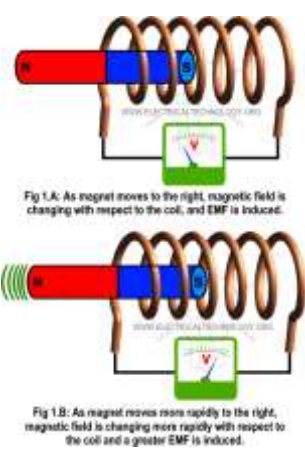
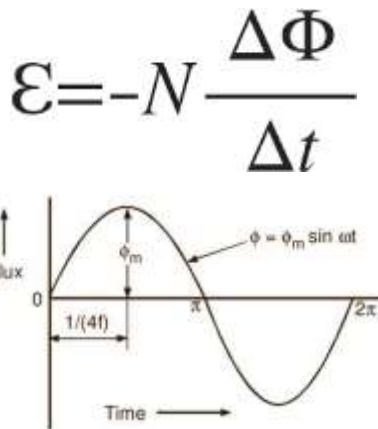
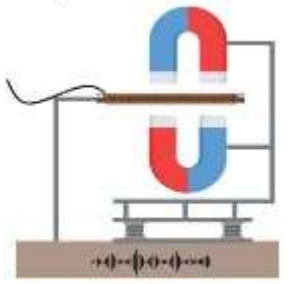


Fig 1.B: As magnet moves more rapidly to the right, magnetic field is changing more rapidly with respect to the coil and a greater EMF is induced.

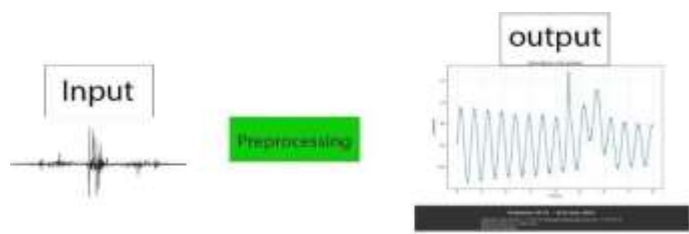
Secondly, **Faraday's Law of Electromagnetic Induction** states that a changing magnetic flux through a coil induces an **EMF**. In our project, a coiled copper wire moves vertically between two **fixed U-shaped magnets**, generating electricity. For sinusoidal motion, the flux varies as “Phi_B = Phi_max x sin(wt)”, producing a sinusoidal EMF. An **EMF vs. time graph** illustrates peaks and zero crossings corresponding to maximum flux changes.



Thirdly, **Springs** beneath the U-shaped magnets cancel horizontal seismic forces, keeping them stationary, while a vertically moving copper coil between the magnets generates electricity via electromagnetic induction. This setup enables safe and efficient energy harvesting from simulated earthquakes.



AI Model Architecture



After inserting the date about the full moon and the animal notification into the AI Model, it has been working well with high accuracy.

Earthquake Simulation System

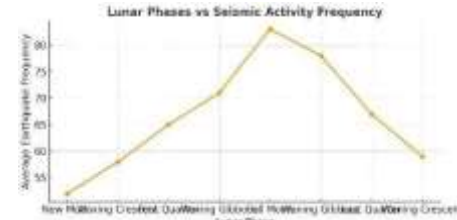
A DC motor connected to two lithium-ion batteries, securely fixed inside a cardboard box, drives a wooden platform to simulate seismic vibrations. The motor's rotation produces controlled oscillations, allowing the system to mimic earthquake motion for testing energy harvesting and damping mechanisms.



AI Model Structure and its Effects

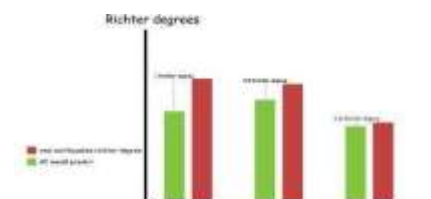


Studies indicate that animals can sense the primary seismic waves (P-waves) before humans, exhibiting unusual behavior prior to a major earthquake (US GS, 2023).

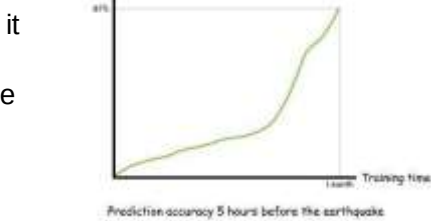


Some studies suggest that large earthquakes are more probable during full moon when tidal stress is maximal.” (Ide et al., 2016).

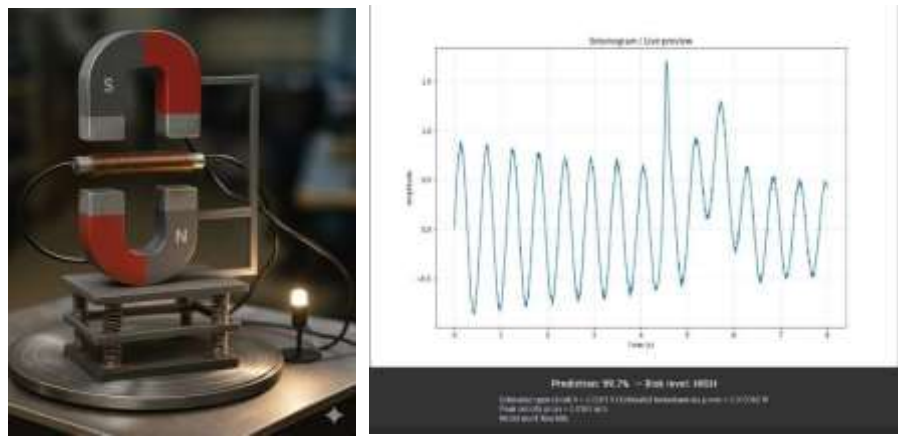
The AI model becomes more accurate as additional seismic data is added. With each new input, it learns patterns more effectively and improves its ability to predict early signs of earthquakes.



The model's accuracy increases as it spends more time training. Longer training periods allow the AI to refine its pattern-recognition ability and reduce errors, resulting in more reliable earthquake predictions.



Graphical Summary Of The Project



Results

Methodology Of The Experiment

How Was It Done?

The project was built by first creating a vibration platform using a **DC motor** connected to **two lithium-ion batteries**, with an eccentric metal weight added to the shaft to simulate earthquake-like horizontal shaking. The entire setup was mounted on a plywood base and stabilized inside a cardboard frame. A spring-based isolation system was then installed under two **U-shaped magnets** to keep them steady while cancelling most horizontal forces. A copper coil wound around an iron core was placed vertically between the magnets. During vibration, the coil moved up and down while the magnets stayed fixed, generating electricity through electromagnetic induction. An **AI model** was also trained on seismic datasets to recognize vibration patterns. Its accuracy improved as more data and training time were added, enhancing its ability to analyze early earthquake signals.



The Results of AI Model

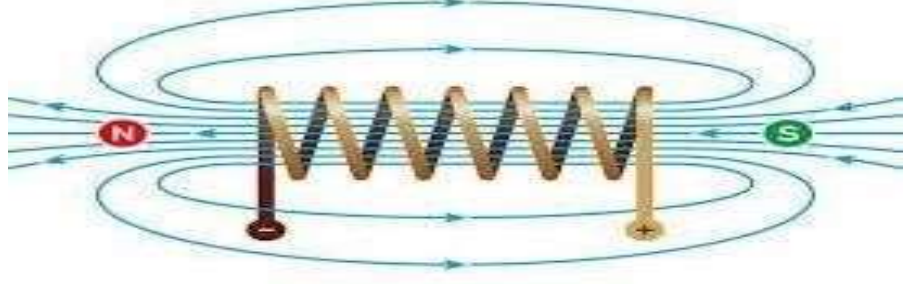
Training Performance s

Table (5), represents the Training performances of AI Model

	Initial vibration	Peak Seismic Oscillation	Damped Oscillation	No-Activity
Sensitivity	99.98	99.89	98.90	99.88
Precision	99.86	99.70	99.87	99.85
Specificity	99.99	99.98	99.98	99.91
Accuracy	99.99	99.97	99.93	99.90
F1-Score	99.92	99.80	98.74	99.86

Final Results Of The Project

The prototype successfully generated enough electricity to light an LED, proving that the system produces a stable and usable output. After improving the coil alignment and motion mechanism, the movement of magnetic flux increased significantly, which led to a noticeable rise in the induced voltage. Multimeter measurements showed a clear increase in output whenever the motion speed increased, confirming that the device follows Faraday's Law. The system operated smoothly across repeated tests with stable readings and no overheating, demonstrating good efficiency and reliability.



Testing Performance s

Table (6), represents the Testing performances of AI Model

	Initial vibration	Peak Seismic Oscillation	Damped Oscillation	No-Activity
Sensitivity	99.14	99.07	98.12	99.30
Precision	99.10	99.07	98.75	99.10
Specificity	99.94	99.94	99.82	99.44
Accuracy	99.90	99.88	99.61	99.59
F1-Score	99.12	99.07	98.44	99.20

Discussion

AI- Model Laws

$$\text{Accuracy} = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Predictions}} = \frac{TP+TN}{TP+TN+FP+FN}$$

$$\text{Sensitivity} = \frac{TP}{TP+FN}$$

$$\text{Specificity} = \frac{TN}{TN+FP}$$

$$\text{Precision} = \frac{TP}{TP+FP}$$

TP (True Positive): Cases correctly predicted as positive.

TN (True Negative): Cases correctly predicted as negative.

FP (False Positive): Cases incorrectly predicted as positive.

FN (False Negative): Cases incorrectly predicted as negative.

Details of LED, Copper



The generated electricity is sufficient to power a small LED, providing a visible demonstration of the system's energy-harvesting capability. The LED lights up reliably during platform vibrations, confirming effective conversion of mechanical motion into usable electrical energy.



A copper wire is tightly coiled around an iron core and positioned vertically between two U-shaped magnets. During platform vibrations, the coil moves up and down while the magnets remain stationary, generating electricity through electromagnetic induction. This setup efficiently converts vertical motion into usable electrical energy.

Conclusions

About **99.79%** Accuracy of the AI Model for earthquake prediction detection by training the Model from QT Data Base.

Real Time Monitoring Enables a real and effective prediction evaluation, providing it before happening

PROBLEM

Technical Limitation

SOLUTION

Serves an accurate real-time prediction for the damaging earthquakes

Coming Plans

The **MPU6050**, can be added to the system to measure vibrations and acceleration in three axes (**X, Y, Z**). It provides precise data on the intensity and direction of seismic-like motion, allowing the AI model to analyze patterns more accurately and optimize energy harvesting from the vertical coil movement.



The electricity generated by the vertical coil is stored in a rechargeable lithium-ion battery instead of powering a simple LED. This allows the system to accumulate energy over time, demonstrating practical energy harvesting and providing a usable power source for later applications.



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